Descriptive Questions – Scenarios in Robotics

Answer each question briefly with justified explanaitons. Each scenario involves specific working environments and challenges. Focus on reasoning when selecting paradigm, actuator, motor, or transmission types.

1) A maintenance robot is designed for university corridors where some floors are made of black marble and others of reflective glass. It must detect walls, avoid reflections, and plan routes efficiently while mopping the surface.
a) Which paradigm (Reactive, Hierarchical, or Hybrid) should be used for this robot and why?
b) Which sensor type (Sonar, Lidar, or Camera) will best support navigation in such environments? Justify.
c) Which transmission system (Gear, Belt, or Chain) should be used for mop arm movement? Why?

2) A hospital service robot is designed to deliver medicines across several departments using elevators. The robot must avoid collisions, wait for doors to open, and adjust paths when humans pass nearby.
a) Which paradigm should be applied for this task and why?
b) Which type of motor (DC, Stepper, or Servo) should be used for precise wheel control during navigation? Explain.
c) Which transmission type (Gear, Belt, or Chain) would be most efficient for smooth, quiet motion? Justify.

3) A logistics robot operates in a factory warehouse, carrying small loads between stations. It sometimes faces uneven floors, dust, and dim lighting conditions.
a) Which control paradigm will ensure stability and reliability in such environments?
b) Which type of sensor (Infrared, Lidar, or Ultrasonic) will help detect obstacles effectively in dusty areas?
c) Which type of actuator (Electric, Pneumatic, or Hydraulic) would suit its load-handling tasks and why?

4) A classroom-assistive robot is designed to pick up small laboratory items like markers and glass beakers from tables and shelves. It must move carefully and place items precisely without causing damage.
a) Which control paradigm (Reactive, Hierarchical, or Hybrid) should be considered for its operation? Why?
b) Which sensor (Camera, Sonar, or Lidar) would help in identifying and positioning objects? Justify.
c) Which motor (DC, Stepper, or Servo) would provide accurate control for the picking mechanism? Explain.

Model Answers (One-Line Each)

1a) Hybrid – combines global path planning with quick response to reflections.
1b) Lidar – stable mapping performance under dark or shiny floors.
1c) Belt drive – smooth and quiet for mop arm motion.

2a) Hybrid – handles dynamic human movement and scheduled navigation.
2b) Servo motor – ensures accurate wheel rotation and speed control.
2c) Gear transmission – durable and efficient for continuous service motion.

3a) Hierarchical – supports controlled task execution in rough environments.
3b) Ultrasonic – works well in dusty and low-light conditions.
3c) Electric actuator – provides reliable force with minimal maintenance.

4a) Hierarchical – allows careful movement and planned grasping.
4b) Camera – detects object shape and location for precise grasping.
4c) Stepper motor – precise angular motion ideal for pick-and-place arms.